Data-Driven Identification of Bifurcation Normal Forms

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1 Introduction

Extracting governing equations from data is a significant challenge in engineering, particularly in complex systems where the underlying dynamics are unknown. In fields such as climate science, financial markets, and neuroscience, vast amounts of data are available, yet the governing equations that describe these processes remain elusive. Often, these equations depend on a few key terms that drive the dynamics, and using sparse regression techniques, it is possible to identify the minimal set of terms required to accurately represent the system. Methods such as Dynamic Mode Decomposition (DMD) and Sparse Identification of Nonlinear Dynamics (SINDy) have proven effective in uncovering these relationships by leveraging sparsity to simplify the identification of governing equations. In this report, we use the SINDy method to analyze data from a typical normal form of an Hopf bifurcation dynamical system, aiming to identify its governing equations from the observed data. This approach highlights the effectiveness of SINDy in capturing the essential dynamics of complex systems by uncovering the minimal set of terms required for accurate representation.

2 Methods Used

The 2-D normal form of the Hopf Bifurcation to be identified :

$$\dot{x} = \mu x + y - x(x^2 + y^2) \tag{1}$$
$$\dot{y} = -x + \mu y - y(x^2 + y^2) \tag{2}$$
$$\dot{\mu} = 0 \tag{3}$$

The normal form of the Hopf Bifurcation used to collect data:

$$\dot{x} = \mu x + y - x(x^2 + y^2) \tag{4}$$
$$\dot{y} = -x + \mu y - y(x^2 + y^2) \tag{5}$$
$$\dot{\mu} = \epsilon \tag{6}$$

This study aimed to investigate the effect of ϵ on the Sparse Identification of Nonlinear Dynamical Systems (SINDy) model and to identify the model which resembles the actual normal form with least error. It was observed that variations in system parameters—such as initial conditions, time span, time step, and the threshold of the STLSQ optimizer—significantly impacted the number of terms and coefficients in the identified system. To determine the most optimal configuration, various models were simulated by adjusting these parameters to minimize error and reduce the number of non-zero terms.

We performed simulations using two values for the perturbation parameter ϵ (0 and 0.1), generating approximately 1.5 million distinct mathematical models each. These models were created by varying initial conditions, time spans, and time steps for each value of ϵ. The initial conditions for the variables x and y were selected from the range $[-1, 1]$, while the initial conditions for the parameter μ were sampled from the interval $[-0.2, 0.6]$. For each model, we randomly chose a time span from the range $[10, 25]$ seconds and a time step from the range $[0.01, 0.05]$ seconds. The SINDy optimizer used the Sequential Thresholded Least Squares (STLSQ) method with a regularization threshold of $\beta = 0.1$ for all models.

3 Results

3.1 Model Selection for $\dot{\mu} = \epsilon = 0$

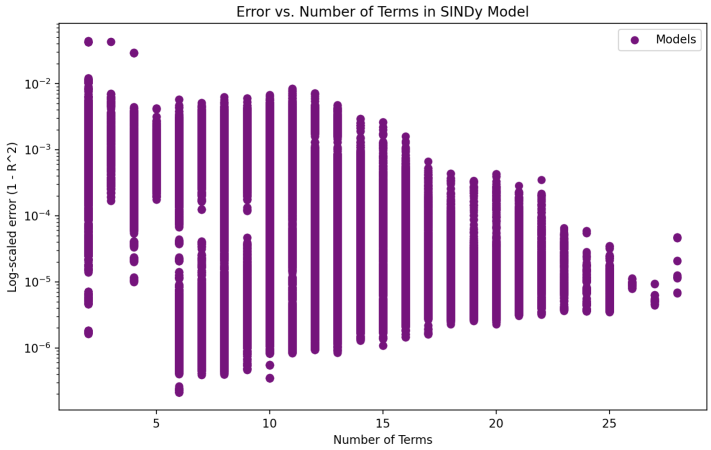


Figure 1: **Effect of** $\mu = 0$. [1].

The identified system with least error when $\dot{\mu} = 0$ is given by the following equations:

$$\dot{x} = -0.201x + 0.962y - 0.192\mu \tag{7}$$
$$\dot{y} = -0.962x - 0.201y + 0.192\mu \tag{8}$$
$$\dot{\mu} = 0 \tag{9}$$

3.2 Model Selection $\dot{\mu} = \epsilon = 0.1$

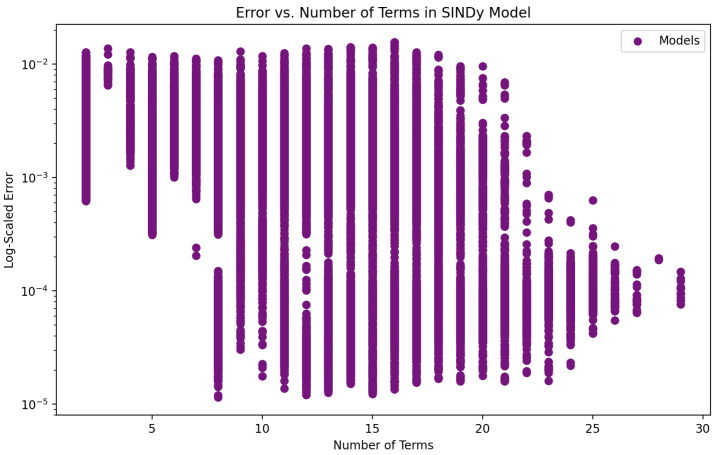


Figure 2: **Effect of** $\mu = 0.1$. [1].

The identified system with least error when $\dot{\mu} = 0.1$ is given by the following equations:

$$\dot{x} = 0.999y + 1.003x\mu - 0.998x^3 - 0.996xy^2 \tag{10}$$
$$\dot{y} = -1.000x + 1.007y\mu - 1.006x^2y - 1.002y^3 \tag{11}$$
$$\dot{\mu} = 0 \tag{12}$$

4 Conclusion

By systematically varying the parameters, we simulated different models and ultimately identified the system that resembles the true system. We observed that as we varied system parameters—namely the initial conditions, time span, and time step—the number of terms in the identified models and the corresponding error changed significantly. While we could not accurately identify the correct system when using $\dot{\mu} = 0$, the correct system was identified with much higher accuracy when $\dot{\mu} = 0.1$ and the threshold value $\beta = 0.1$. We believe this is because when $\dot{\mu} = 0$, the calculations occur within the null space, yielding unsatisfactory results. However, when $\dot{\mu} = 0.1$, which equals the threshold value β, the system does not remain in the null space, but the coefficient for μ is masked to zero. If ϵ exceeds the threshold β, the correct system would not be identified.

References

[1] S. L. Brunton, J. L. Proctor, and J. N. Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the national academy of sciences*, 113(15):3932–3937, 2016.